

Research

Green creativity as a mediator between green human resource management practices and sustainable development: insight from Saudi Arabia

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Abstract

Green Human Resource Management (GHRM) has emerged as a key strategic approach for aligning organizational practices with environmental sustainability goals. While prior studies have primarily examined the direct influence of GHRM on sustainable development (SD), limited attention has been given to the mediating role of green creativity (GC), particularly in emerging economies like Saudi Arabia. This study aims to investigate the relationship between GHRM practices and SD, focusing on the mediating effect of GC. Data were collected from 250 employees in diverse private-sector organizations and analyzed using Structural Equation Modeling via Mplus. The findings reveal that GHRM practices positively influence both GC and SD. More importantly, GC significantly mediates the relationship between GHRM and SD. This study highlights the critical role of fostering employee creativity through GHRM practices, such as training, performance management, and employee involvement, to achieve sustainability outcomes. Theoretically, the study extends the application of the theory of planned behavior by positioning GC as a key behavioral mechanism linking HR practices to sustainability outcomes. It also offers practical insights for organizations seeking to enhance their sustainability performance by integrating environmentally focused HRM strategies and cultivating a culture of innovation, in line with the sustainability objectives outlined in Vision 2030. *Clinical trial number*: "Not applicable."

Keywords Green human resource management · Green creativity · Sustainable development · Saudi private sector · Structural equation modeling

1 Introduction

In the face of escalating environmental challenges, sustainable development (SD) has become an imperative for governments, organizations, and societies worldwide. The United Nations' Sustainable Development Goals (SDGs), adopted in 2015, provide a strategic framework for balancing economic growth, social equity, and environmental preservation [50]. Among the many actors in the sustainability transition, organizations, particularly those in the private sector, are increasingly recognized as pivotal agents due to their operational scale and resource influence. Consequently, businesses are under growing pressure to integrate environmental and social considerations into their strategic frameworks, not only to satisfy regulatory requirements but also to enhance competitiveness, employee engagement, and brand reputation [61, 70].

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Amid this shift, green human resource management (GHRM) has emerged as a key strategic approach to embedding sustainability into the fabric of organizational culture and processes. GHRM involves aligning traditional human resource practices, such as recruitment, training, performance management, and employee engagement, with environmental objectives [46, 66]. These practices aim to create a workforce that is not only environmentally conscious but also capable of initiating and supporting sustainable innovations [3, 34]. By promoting green values, encouraging pro-environmental behaviors, and facilitating eco-innovation, GHRM contributes directly and indirectly to SD [38].

Despite a growing body of research highlighting the importance of GHRM in advancing SD [18, 68], significant gaps remain in understanding the mechanisms through which GHRM contributes to SD [11, 51]. First, many empirical studies have been conducted in developed economies [53, 54], with relatively little attention given to emerging economies like Saudi Arabia. This contextual gap is particularly important in light of the ongoing economic transformation in Saudi Arabia under its Vision 2030 strategy. Historically dependent on oil, Saudi Arabia is now actively pursuing economic diversification under Vision 2030 [8], with aims to increase the private sector's contribution to GDP from 40 to 65% and reduce reliance on oil revenues [6, 20]. As part of this strategic shift, Saudi Arabia has committed to promoting environmental sustainability, green innovation, and corporate social responsibility across all sectors [7]. However, it still faces serious environmental challenges, where Saudi Arabia ranks among the top nations in per capita CO₂ emissions, with an annual output of approximately 19 metric tons [14], alongside challenges such as water scarcity, rising temperatures, and growing energy demands [8]. These pressing issues highlight the need for context-specific organizational strategies that align with both national and international sustainability objectives, underscoring the importance of in-depth, context-specific investigations on sustainability practices within organizations.

This contextual gap is particularly important in light of Saudi Vision 2030 goals, which require tailored organizational strategies that support green transformation and sustainable innovation [7]. The unique economic structure of Saudi Arabia, characterized by its reliance on oil, emerging private sector, and evolving regulatory landscape, necessitates context-specific investigations into how sustainability is operationalized within organizations [8].

Second, while prior research has established that GHRM can positively impact SD, the underlying mechanisms through which this occurs remain unclear. Much of the existing work treats the GHRM–SD link as direct, ignoring potential mediating and/or moderating factors that may explain how GHRM shapes sustainability outcomes. Moreover, existing findings are mixed, with some studies reporting weak or even negative links between GHRM practices and corporate sustainability (e.g., Elshaer et al. 2021; Niazi et al. 2023), highlighting the need for more in-depth investigation. To address this gap, the present study aims to explore the role of green creativity (GC) as a mediating factor in the GHRM–SD relationship, based on the premise that GC represents an important behavioral outcome of GHRM efforts and a potential enabler of sustainable development within organizations [33, 74]. We conceptualize GC as a key behavioral mediator between GHRM and sustainable development, based on the theory of planned behavior (TPB) [5]. TPB posits that behavior is shaped by attitudes, subjective norms, and perceived behavioral control. In an organizational context, GHRM practices influence these psychological precursors by fostering positive environmental attitudes, reinforcing sustainability-focused norms, and empowering employees with the resources and autonomy necessary for green action. These factors, in turn, encourage employees to engage in GC, which can lead to innovative, environmentally friendly outcomes that support broader sustainability goals [56, 65].

In light of the gaps identified in the literature above, this study seeks to investigate the following questions:

RQ1: How do GHRM practices influence SD?

RQ2: Does GC mediate the GHRM and SD relationship?

By answering these questions, the contribution of this study is twofold. First, it deepens our understanding of how GHRM practices support SD within the context of an emerging economy, Saudi Arabia, where this issue is particularly significant. The country's Vision 2030 sets forth an ambitious agenda aimed at economic diversification, innovation, and environmental responsibility. As Saudi Arabia shifts from an oil-based economy to one centered on knowledge and innovation, there is growing attention on organizational strategies that promote environmental responsibility and drive innovative sustainability efforts [8]. Therefore, examining the behavioral and organizational drivers of sustainability, such as GHRM and green creativity, is vital to advancing this national vision. Second, it explores GC as a key mediating mechanism, offering a novel behavioral pathway that links GHRM practices to sustainability outcomes through the lens of the TPB. By doing so, the study provides actionable insights for HR practitioners and policymakers aiming to align human resource strategies with environmental goals, thereby supporting the sustainability agenda embedded in Saudi Vision 2030.

The remainder of this study is organized as follows: Sect. 2 presents a comprehensive review of the relevant literature and theoretical foundation, culminating in the development of the study's hypotheses. Section 3 outlines the research methodology. Section 4 details the results of the empirical analysis using structural equation modeling. Section 5

discusses the key findings, while Sect. 6 covers theoretical and practical implications, acknowledging limitations, and suggesting directions for future research.

2 Literature review and hypotheses formulation

2.1 Planned behavior theory

Guided by TPB, this study developed a robust framework for understanding how GHRM practices influence SD through the mediating role of GC. According to TPB, individual behavior is guided by three psychological components: attitudes, subjective norms, and perceived behavioral control, which together shape behavioral intentions and actual actions [5]. In the context of GHRM, practices such as green training, performance management, and employee involvement enhance pro-environmental attitudes, establish supportive organizational norms, and increase employees' confidence in their ability to contribute to environmental goals [62, 66]. These factors foster an environment that encourages GC, which is viewed as the generation of novel and valuable ideas aimed at improving environmental performance [28]. GC acts as a behavioral outcome of the cognitive and motivational processes influenced by GHRM, thereby serving as a critical link between HRM practices and sustainable organizational outcomes. As such, TPB supports the conceptualization of GC as a mediator by highlighting the psychological mechanisms through which GHRM practices stimulate creative, sustainability-oriented behaviors that ultimately contribute to SD [56, 65].

2.2 Green human resource management practices and sustainable development

Recently, GHRM has been recognized as a strategic business approach, embedding environmental considerations into the core of HRM processes [74]. GHRM plays a crucial role in shaping organizational efforts toward sustainable development (SD). Rooted in the integration of environmental concerns into traditional human resource practices, GHRM encompasses a spectrum of activities, including green recruitment, sustainability-focused training, environmentally oriented performance appraisals, and the incentivization of eco-friendly behaviors [17]. These initiatives not only promote responsible environmental actions among employees but also contribute to reduced carbon footprints, enhanced productivity, and improved employee engagement and motivation [58, 59].

The adoption of GHRM enables organizations to operationalize sustainability goals by embedding environmental values within the organizational culture. As Al-Shammari et al. [11] observed, the integration of sustainability into strategic management practices has been a key driver for the evolution of GHRM, particularly in the Saudi context, where sustainability is gaining traction as a competitive advantage. Singh et al. [69] reinforced this by highlighting how aligning HRM with sustainability leads to improvements in talent retention, organizational performance, and corporate social responsibility outcomes. Liu et al. [51] further noted that GHRM facilitates the transition toward sustainable business models, positioning it as an enabler of long-term environmental and economic objectives.

Based on TPB, which posits that individual behavior is influenced by attitudes, subjective norms, and perceived behavioral control [5], GHRM plays a role in shaping each of these elements. For instance, environmentally focused training fosters pro-environmental attitudes among employees, while green performance appraisals and leadership commitment cultivate supportive social norms. Moreover, empowering employees with the resources and autonomy to engage in sustainable behaviors enhances their perceived behavioral control. These psychological mechanisms, as Chen et al. [27] demonstrated in the context of the oil and mining industries, significantly impact employees' intentions and actions toward sustainable practices.

In this regard, Ahmed et al. [4] stressed that achieving corporate sustainability requires collective stakeholder engagement, including employees, management, and customers, which can be facilitated through GHRM practices. By shaping employees' pro-environmental attitudes, reinforcing sustainability norms, and increasing behavioral control, GHRM strengthens their intentions to engage in environmentally responsible actions, which are critical drivers of sustainable development [7]. These practices become embedded in the organizational fabric, influencing daily operations, decision-making processes, and strategic objectives. As organizations increasingly integrate sustainability into their strategic management agendas [11, 69], GHRM emerges as a critical enabler of this transformation, contributing not only to improved environmental performance but also to long-term organizational resilience and stakeholder trust [51]. Based on the above, we propose the following hypothesis:

Hypothesis (H1): GHRM practices are positively associated with SD.

2.3 Green human resource management practices with green creativity

GHRM has emerged as a vital organizational approach that integrates environmental sustainability into human resource practices. Its significance lies not only in promoting environmental responsibility but also in fostering innovative behaviors such as GC [10], which refers to the generation of novel and valuable ideas aimed at improving environmental sustainability within organizational settings [41]. GC enables employees to imagine and implement ways to minimize resource usage and improve the organization's environmental footprint, thereby offering a dual advantage—contributing to both organizational innovation and ecological outcomes [74]. It plays a critical role in helping organizations explore new opportunities for sustainable products, services, and business models that can lead to competitive advantage in the context of growing environmental awareness [33].

A core argument for the link between GHRM and GC is grounded in TPB, which posits that behavioral intentions, and ultimately behaviors, are shaped by attitudes, subjective norms, and perceived behavioral control [5]. In the context of GHRM, these three antecedents are systematically supported through GHRM practices. First, GHRM practices such as green training and development are instrumental in cultivating positive attitudes toward environmental innovation by equipping employees with the necessary knowledge and skills to engage in green thinking [62]. Second, the creation of a workplace culture that promotes environmental values helps establish subjective norms, where pro-environmental behavior is expected and socially reinforced within teams and departments. Third, GHRM enhances perceived behavioral control by offering autonomy, decision-making freedom, and participative opportunities, thus empowering employees to act on their green ideas with confidence [66, 68].

Furthermore, GHRM facilitates employee engagement in environmental initiatives by providing the necessary resources, platforms, and motivational incentives [7]. Green performance evaluations, eco-based rewards, and participatory decision-making processes serve to inspire and reinforce employees' commitment to sustainability goals [55, 60]. As a result, employees internalize the importance of their role in environmental improvement and are more likely to exhibit proactive behaviors such as green creativity. In this way, GHRM contributes to shaping a psychologically empowered workforce that is both motivated and capable of generating innovative environmental solutions [43].

In addition, empirical evidence (e.g., [35, 40]) provides support for the notion that GHRM practices have a positive association with GC. By fostering a culture of sustainability, GHRM not only improves organizational alignment with environmental goals but also encourages the emergence of innovative solutions that are essential for meeting sustainability targets [28, 41]. Therefore, based on the theoretical support provided by TPB and the empirical evidence linking GHRM practices to creative environmental outcomes, it is posited that GHRM practices play a foundational role in fostering GC among employees. Hence:

Hypothesis (H2): Green Human Resource Management practices are positively associated with green creativity.

2.4 Mediating role of green creativity

While existing literature has established a positive association between GHRM and sustainable development (e.g., [1, 2, 34]), the underlying mechanisms through which this relationship occurs remain limited. Drawing on TPB, it becomes evident how GHRM may indirectly influence sustainable development outcomes by first shaping employees' psychological intentions toward environmental action [36]. In the context of GHRM, environmentally oriented training and leadership practices enhance employees' pro-environmental attitudes, establish sustainability as a social norm, and empower individuals with the autonomy and competence needed to act on green intentions [7]. These psychological factors, in turn, create a fertile ground for GC to emerge as employees are motivated and equipped to develop innovative, sustainability-driven ideas and practices [28, 62]. Therefore, GC can be viewed as the behavioral output of the cognitive and motivational processes triggered by GHRM [4, 31, 45].

Moreover, GC plays a critical role in translating green intentions into impactful sustainability outcomes [52]. Sustainable development often requires creative problem-solving to address complex environmental challenges, including resource scarcity, waste reduction, and energy efficiency [15, 19]. Employees exhibiting GC are more likely to propose new green products, redesign processes for better ecological outcomes, and engage in eco-innovation initiatives, all of which contribute directly to the organization's sustainable development agenda [33, 56]. From this perspective, GC serves as a behavioral conduit that channels the motivational influence of GHRM into tangible sustainability outputs.

Without this creative engagement, the potential of GHRM to drive sustainability may remain limited to policy compliance rather than transformative change.

Empirical findings further substantiate the mediating role of GC. For example, Arslan et al. [16] have shown that GC plays as a mediator between environmental consciousness and employee energy efficiency. Riva et al. [67] also revealed that GC is not only a desirable outcome of green management practices but also a mechanism through which HR-driven environmental initiatives (green transformational leadership) translate into broader sustainability outcomes. Based on the above theoretical rationale and empirical support, we propose that:

Hypothesis (H3): Green creativity is positively associated with sustainable development.

Hypothesis (H4): Green creativity mediates the relationship between Green Human Resource Management and sustainable development.

As discussed, Fig. 1 provides a visual representation of the study's model.

3 Methods

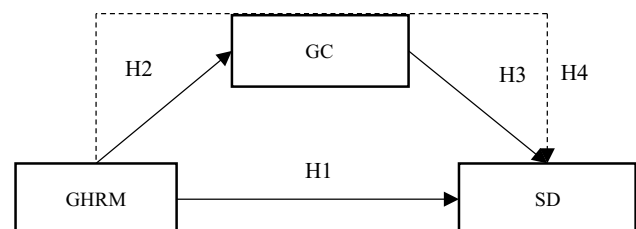
3.1 Sample and data collection

To fulfill the study's objectives, a quantitative research design was adopted, utilizing a structured questionnaire to collect data from employees working in Saudi Arabia's private sector. This sector was chosen for several strategic and contextual reasons. Under Saudi Arabia's Vision 2030, the private sector is expected to play a central role in national development by increasing its contribution to GDP from 40 to 65% [6, 20]. In this transformative context, private organizations are increasingly expected to adopt environmentally responsible strategies, making them a vital domain for exploring the impact of GHRM on sustainability outcomes. Additionally, the private sector in Saudi Arabia is notably diverse, encompassing a broad spectrum of industries and firm sizes. To reflect this diversity, our study included employees from a range of small, medium, and large enterprises operating in key industries such as manufacturing, services, construction, and retail. This inclusive approach is consistent with prior studies emphasizing the importance of contextual and organizational heterogeneity in sustainability research [8, 18].

Due to the practical constraints of reaching a probabilistic sample across multiple industries and regions, we employed a voluntary (non-probability) sampling strategy, wherein participants opted into the study of their own accord. This method is suitable in organizational research when time, access, or logistical resources are limited [47, 48]. To mitigate potential selection bias and improve the representativeness of the data, we ensured diversity across job roles, organizational sizes, and industry sectors, following established guidelines for enhancing external validity in non-probabilistic research [63]. Moreover, confidentiality and anonymity were prioritized to encourage candid participation. All questionnaire items were neutrally phrased to avoid leading responses, and informed consent was obtained digitally before participation began. The survey was distributed online using Google Forms, which enabled access to respondents across different regions in Saudi Arabia over a 7-month period, from September 2023 to March 2024.

To determine the appropriate sample size, the study followed a subject-to-item ratio [71], following Bentler and Chou's [24] guidelines that a minimum of five respondents be allocated per parameter. A total of 263 responses were initially collected. After data screening and cleaning for completeness and consistency, 250 valid responses were retained for analysis, exceeding the initial target of at least 225 participants. As a result, the response rate is 95% of the total sample, which is much higher compared to the response rates reported in many studies in the field of management [9, 12]. To determine the statistical power and accurately estimate the minimum required sample size,

Fig. 1 Study model



G*Power analysis was utilized [37]. The analysis indicated that the sample size of 68 cases is adequate. Therefore, our sample of 250 cases significantly surpassed this threshold, demonstrating that our sample size was adequate and appropriate for yielding reliable and valid results. Table 1 presents the characteristics of the sample.

3.2 Measures

For this study, a questionnaire tool was used for data collection, which was chosen for its efficiency in gathering substantial data within a reasonable timeframe and budget [44]. After developing the questionnaire and before distribution, a two-stage pretest was conducted to minimize potential bias in data collection and guarantee the scale's validity and reliability. In the first stage, four academics reviewed the wording and purpose of the questions. In the second stage, following the guidance of Perneger et al. (2015), we solicited feedback from ten participants regarding the question format, wording, length, and sequence. Based on their feedback, we made only minor adjustments, primarily rearranging the order of some questions, before finalizing the questionnaire for data collection. This approach ensured both content validity and comprehensive representation of each construct.

Overall, the measures of this study were extracted from previous relevant research. GHRM was measured with a scale adapted from Tang et al. [72], which evaluate green training, performance management, compensation and incentives, and employee involvement. The scale demonstrated strong internal consistency with a Cronbach's alpha of 0.93 in this study.

While GC was evaluated using a six-item instrument adapted from Chen and Chang [28], scored on a five-point Likert scale (1 = strongly disagree, 5 = strongly agree). One illustrative statement from the scale reads, "Team members engaged in green product development propose innovative ways to accomplish environmental objectives." In the current investigation, this measure demonstrated strong reliability, as indicated by Cronbach's alpha value of 0.85.

Finally, SD was evaluated as a multi-dimensional instrument, with a scale adapted from Chow and Chen [30] scored on a five-point Likert format (1 = small extent, 5 = large extent). The scale exhibited a high level of reliability in this research, reflected by a Cronbach's alpha of 0.92. Appendix 1 demonstrates all items used to measure the study's constructs.

3.3 Common method bias

To assess potential common method bias (CMB), Harman's single-factor test was employed, following the approach outlined by Podsakoff and Organ [64]. The analysis showed that one factor accounted for 27.566% of the variance, which

Table 1 Demographic characteristics of the participants

Variables	Category	Frequency	Percentage
Gender	Male	137	54.8
	Female	113	45.2
Marital Status	Married	120	48
	Unmarried	130	52
Age (years)	Under 30	80	32
	30–34	70	28
	35–39	5	2
	40–44	45	18
	More than 45	50	20
Educational level	High school	26	10.4
	Diploma	19	7.6
	Bachelor	180	72
	Graduate	25	10
Total Experience (years)	Less than 3	4	1.6
	3–10	146	58.4
	10–15	5	2
	More than 15	95	38

is well under the 50% threshold. This indicates that CMB is unlikely to pose significant concern in this study. More comprehensive results from this test are presented in Table 2.

3.4 Data analysis techniques

For analysis, SPSS (version 23) was utilized for assessing reliability, generating descriptive statistics, and conducting correlation analyses. The measurement model was evaluated using Mplus (version 7.1), after which path analysis was performed through Structural Equation Modeling (SEM). All constructs were assessed on a five-point Likert scale. Given that responses were categorical, the Weighted Least Squares Means and Variance-adjusted (WLSMV) estimator was applied during path analysis and to verify measurement model fit, offering robustness against non-normal data distributions and outliers.

Mplus software (version 7.1) was employed for data analysis because it is in line with the nature of this study, which is theory-driven and confirmatory, aimed at empirically testing the mediating role of GC between GHRM and SD. Given this context, a covariance-based structural equation modeling (CB-SEM) approach was deemed more appropriate than partial least squares SEM (PLS-SEM), which is better suited for exploratory or prediction-oriented studies [39]. Among CB-SEM software options, Mplus offers specific advantages. Firstly, it supports the WLSMV estimator, which is particularly robust when dealing with ordinal data such as Likert-scale responses [22, 26]. This estimator does not assume multivariate normality and provides more accurate parameter estimates and standard errors than maximum likelihood methods when data are non-normally distributed [57]. Moreover, Mplus allows for sophisticated modeling of mediation and indirect effects, including bootstrapping and bias-corrected confidence intervals, which were crucial for our study's mediation analysis involving GC [29]. Moreover, Mplus supports second-order factor models and model constraints, allowing for greater flexibility and precision in testing our theoretical model [49].

In contrast, while PLS-SEM is advantageous for smaller samples and complex exploratory models, it lacks the capability to provide global goodness-of-fit indices, which are necessary for assessing model fit in confirmatory studies [39]. Similarly, although AMOS is capable of CB-SEM, they do not offer the same level of flexibility in handling categorical data and advanced mediation analysis as Mplus. Based on these methodological strengths, Mplus was chosen as the most appropriate tool for achieving the analytical rigor and model accuracy required for this study.

4 Results

4.1 Measurement model evaluation

4.1.1 Convergent and discriminant validity

Convergent validity was assessed by examining the Average Variance Extracted (AVE) for each construct. In all cases, AVE values surpassed 0.50, suggesting that the constructs captured adequate variance in relation to their respective indicators Churchill Jr, [32]. In addition, the factor loadings were greater than 0.60, indicating the validity of the measurement model [21], as shown in Appendix 1. Furthermore, each construct's Composite Reliability (CR) exceeded 0.70, reinforcing the reliability and internal consistency of the measures. Discriminant validity was also supported, as the values of the AVEs for the constructs were greater than the correlations among the latent variables. A summary of these findings is provided in Table 3.

4.1.2 CFA analysis

Confirmatory Factor Analysis (CFA) was used to assess the measurement models. Given that certain constructs in this study were second-order, all first-order factors were initially combined into a single measurement model to validate their reliability and validity. Once those properties were confirmed, another CFA was conducted to evaluate the second-order structures. Specifically, green recruitment and selection, training, performance management, compensation and rewards, and employee involvement collectively made up the second-order construct of GHRM. Likewise, social development, economic development, and environmental development formed the second-order

Table 2 Harman's Single-Factor Test

Total variance explained					
Component	Initial eigenvalues		Extraction sums of squared loadings		
	Total	% of variance	Cumulative %	Total	% of variance
1	12.837	29.174	29.174	12.129	27.566
					27.566

Table 3 Composite reliability, discriminant of 2nd-order constructs, convergent validity

Construct	GC	GHRM	SD	CR	AVE
GC	0.709			0.859	0.503
GHRM	0.454	0.791		0.893	0.625
SD	0.581	0.596	0.736	0.778	0.542

The bold values in the diagonal are square roots of AVEs of respective constructs. GC=green creativity, GHRM=green human resource management, SD=sustainable development

construct of SD. Model fit indices (GOF) were then computed and compared to the guidelines set forth by Hair et al. [39], that is, $X^2/df < 3$, $CFI > 0.95$, $TLI > 0.95$, $RMSEA < 0.08$, and $SRMR < 0.08$. The measurement models showed an excellent model fit, as shown in Table 4.

4.2 Structural model evaluation

4.2.1 Structural model fit

When evaluating the model's goodness-of-fit, a range of indices was applied and compared against recognized standards for the structural model. The Comparative Fit Index (CFI) was observed at 0.954, the Root Mean Square Error of Approximation (RMSEA) stood at 0.374, and the Tucker-Lewis Index (TLI) reached 0.95—all of which point to an excellent model fit. As illustrated in Table 5, these outcomes confirm that the hypothesized links among the variables are robustly upheld by the data [23, 25, 42].

4.2.2 Structural model results

To examine the relationships between the variables of the study, the hypotheses were tested. As presented in Table 6, the results indicate that GHRM practices have a significant and positive effect on GC and SD. Therefore, H1 and H2 were accepted. In addition, GC has a significant and positive effect on SD. H3 was thus supported. Additionally, GHRM practices positively influence SD through GC. As a result, H4 was supported. Table 6 and Fig. 2 shows the result of Hypothesis testing, respectively.

5 Discussion

Guided by TPB, this study sought to investigate the mechanisms through which GHRM practices influence SD. First, the findings confirm the direct positive relationship between GHRM practices and SD. This indicates that environmentally oriented HR strategies, such as green recruitment, training, performance appraisals, and employee involvement, contribute

Table 4 Model Fit indices of measurement models

Models	X2	df	CFI	TLI	SRMR	RMSEA	LL	UL
1st-order measurement model	858.60	666	0.962	0.958	0.046	0.034	0.027	0.040
2nd order measurement model	926.07	691	0.954	0.951	0.060	0.037	0.030	0.043

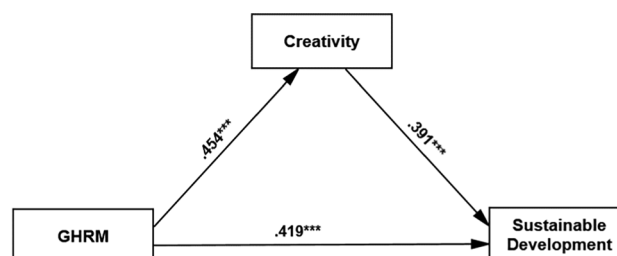
RMSEA=Root Mean Square Error of Approximation, SRMR=Standardized Root Mean Square Residual, CFI=Comparative Fit Index, TLI=Tucker-Lewis Index, CI=Confidence Interval, LL=Lower Limit, UL=Upper Limit

Table 5 GOF indices of Structural model

Models	X2	df	CFI	TLI	SRMR	RMSEA	LL	UL
Structural Model 1	926.07	691	0.954	0.951	0.060	0.037	0.030	0.043

Table 6 Hypothesis testing results

Relationship	Estimates	SE	Results
GHRM → SD	0.454***	0.062	Supported
GHRM → GC	0.419***	0.074	Supported
GC → SD	0.391***	0.074	Supported
GHRM → GC → SD	0.178***	0.040	Supported

*** = relationship is significant at $p < 0.001$ **Fig. 2** Structural model and standardized estimate.
*** $p < 0.001$ 

significantly to achieving sustainability goals. This supports the assertions of Renwick et al. [66], who argued that embedding green values within HR functions fosters an environmentally responsible organizational culture. Our findings are also aligned with more recent studies such as those by Al-Swidi et al. [13], Al-Shammari et al. [11] and Vázquez-Brust et al. [73], which emphasized GHRM's role in promoting ecological performance. However, some earlier research (e.g., Elshaer et al. 2021, Niazi et al. 2023) reported inconsistent or weak effects, especially in contexts where environmental strategies are seen as symbolic rather than operational. In contrast, our findings suggest that in the context of Saudi Arabia, where Vision 2030 is actively pushing sustainability initiatives, GHRM is being operationalized in meaningful ways that support organizational and national sustainability objectives.

Our findings also revealed a positive and significant relationship between GHRM and GC, which aligns with TPB, as GHRM positively shapes employee attitudes, perceived norms, and behavioral control, leading to creative, pro-environmental outcomes. This finding is consistent with Hameed et al. [40] and Pham et al. [61], who found that GHRM practices (green training and empowerment) foster creative thinking and ecological problem-solving among employees.

Additionally, our results validate GC as a significant predictor of SD, highlighting that GC is not only a desirable organizational trait but a vital mechanism for achieving sustainability. This is consistent with prior research by Mitchell and Walinga [56] and Zahrani [74], which emphasized the importance of creative problem-solving in addressing complex environmental challenges. Employees who are encouraged to think creatively about environmental issues are more likely to design eco-innovative products, suggest green processes, and engage in environmentally conscious behaviors that benefit the organization's sustainability efforts.

Most critically, our study confirms the mediating role of GC between GHRM and SD, thereby providing empirical evidence for a behavioral pathway grounded in TPB. While prior studies have suggested the potential of this pathway (e.g., [45, 67]), few have tested it within a structural model that incorporates contextual nuances. This mediating effect highlights that simply adopting GHRM practices may not be enough, where organizations must actively cultivate environments that enable employees to express and apply their creative capabilities to achieve sustainability goals.

6 Conclusions

6.1 Theoretical implications

Theoretically, this study has several implications. First, this study makes a significant contribution to the body of knowledge on GHRM and SD by introducing GC as a mediating element in the GHRM–SD link. Although earlier research primarily concentrated on the direct association between GHRM and sustainability outcomes, this work offers empirical evidence that underscores the importance of innovation and creativity within HRM processes. In doing so, it enriches discussions surrounding the role of HRM in fulfilling the United Nations SDGs. Second, this study advances TPB by demonstrating its applicability in the context of organizational sustainability, particularly

within GHRM practices. While TPB has traditionally been employed to explain individual behavior in environmental or consumer environments, our study expands this scope by illustrating how GHRM practices influence the three core components of TPB (attitudes, subjective norms, and perceived behavioral control), which serve as a behavioral manifestation of employees' pro-environmental intentions. By applying TPB, this study adds theoretical depth by revealing a psychological-behavioral mechanism that connects GHRM practices to SD. By doing so, it bridges the gap between employee-level cognition and macro-level sustainability outcomes, a connection that has been relatively underexplored in TPB-based sustainability research. The framework presented here also sets the stage for future research to examine other behavioral mediators (e.g., environmental identity, green organizational citizenship behavior) that could further elaborate the GHRM–SD relationship within TPB. Lastly, this study is grounded in the Saudi Arabian private sector context; however, the theoretical mechanisms derived from TPB and tested using SEM are applicable across other emerging economies facing similar sustainability and innovation imperatives. As such, our model provides a transferable framework that can be tested in different cultural, economic, and regulatory environments to assess its broader applicability and theoretical consistency.

6.2 Managerial implications

From a practical standpoint, the findings of this study offer several targeted recommendations for managers operating in the Saudi Arabian private sector, particularly in light of the country's Vision 2030 sustainability agenda. The results highlight the critical importance of embedding GHRM practices into daily operations to foster GC and, in turn, drive SD. To translate these findings into actionable strategies, organizations should begin by integrating environmental content into their employee training and development programs. This can be achieved by embedding sustainability case studies and locally relevant examples, such as those from the construction, energy, and retail sectors, into training modules, ensuring that employees continuously develop their capabilities to support green innovation. In addition, managers should consider revising performance appraisal systems to include clear indicators that capture eco-friendly behaviors and outcomes. Incentive structures can be modified to reward not only individual but also team contributions to green initiatives, thereby reinforcing a culture that values sustainability. In the realm of recruitment, organizations are encouraged to prioritize candidates who demonstrate a commitment to environmental values. This can be achieved by updating job descriptions to reflect environmental responsibilities and incorporating sustainability-oriented questions into the interview process. Moreover, managers should create an organizational environment that actively encourages GC by establishing formal platforms, such as internal suggestion systems or green idea contests, through which employees can propose innovative solutions to environmental challenges. Offering small budgets or pilot opportunities for selected ideas can foster a sense of ownership and encourage experimentation. Furthermore, managers can support cross-functional collaboration by forming green task forces that bring together staff from HR, operations, R&D, and other departments to co-develop practical, sustainability-driven initiatives. Lastly, it is essential that these GHRM initiatives are not implemented in isolation but are explicitly aligned with the broader objectives of Vision 2030, as this can strengthen organizational legitimacy and enhance top management support for environmental strategies. By doing so, organizations can ensure that their internal sustainability practices contribute meaningfully to national priorities while also enhancing their competitiveness, stakeholder trust, and long-term resilience in an increasingly eco-conscious business environment.

6.3 Limitations and future work

While this study contributes meaningful perspectives, certain constraints merit acknowledgment. The primary limitation stems from its exclusive focus on the Saudi context, which may limit broader relevance to dissimilar cultural or organizational settings. Future research could test this framework in diverse geographic or institutional environments to evaluate the consistency of relationships among GHRM practices, GC, and sustainability outcomes. Second, the study employed a non-probability voluntary sampling technique, which may restrict the generalizability of the findings beyond the specific context of Saudi Arabia's private sector. Although efforts were made to enhance sample

diversity across industries, job roles, and organization sizes, future research could benefit from employing stratified random sampling to improve external validity and ensure broader representativeness. Third, data collected through self-reported questionnaires, which may be subject to response biases such as social desirability or consistency artifacts. While Harman's single-factor test revealed that common method bias was not a serious concern in this study, future research could adopt longitudinal or multi-source data collection strategies (e.g., supervisor or peer evaluations) to minimize potential bias and strengthen the robustness of results. Fourth, the cross-sectional nature of the study limits the ability to make definitive causal inferences. Although SEM using Mplus enabled rigorous mediation analysis, longitudinal designs would be more appropriate for examining the temporal dynamics between GHRM practices, green creativity, and sustainability outcomes. Fifth, while GC was incorporated as a key mediator in the GHRM–SD relationship, other potential mediators, such as environmental identity, green organizational citizenship behavior, or pro-environmental knowledge, were not explored. Future studies should investigate these alternative or complementary pathways to provide a more comprehensive understanding of how GHRM practices contribute to sustainable organizational performance. Moreover, external contextual factors may play a critical role in determining the effectiveness of GHRM initiatives. Future studies are encouraged to examine how factors such as regulatory frameworks, financial incentives, and technological developments might influence, or potentially moderate, the relationship between GHRM and SD. Such investigations could offer a more nuanced understanding of the dynamics involved. Lastly, while this study applied the TPB to explore the pathway from GHRM to sustainability via GC, future research could benefit from utilizing alternative theoretical lenses, such as the natural resource-based view, institutional theory, or stakeholder theory, to uncover new perspectives on how organizations respond to sustainability imperatives.

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Data availability Data sharing is not applicable. The data are not publicly available because of participants' privacy.

Declarations

Ethics approval and consent to participate “The study was conducted in accordance with the Declaration of Helsinki, and approved by the Institutional Review Board (Ethics Committee) of King Saud University (No: KSU-HE-24-162), Date: 27/2/2024. The approval was granted after the committee reviewed the research proposal and questionnaire items.”

Consent to participate “Participants were given written informed consent forms that outlined the study's nature and purpose, as well as their rights and responsibilities. Since data was collected through online survey, this served as written authorization. Participants were assured that they could withdraw from the study at any time without any repercussions and that their participation in it was entirely voluntary.”

Consent for publication “Not applicable.”

Competing interests The authors declare no competing interests.

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Appendix 1. Questionnaire items

Construct	Items	Loadings
Green Human Resource Management	Our firm attracts green HRM conscious' candidates who use green HRM criteria to select organizations.	.717
	Our firm uses Green HRM employer branding to attract green HRM conscious employees.	.777
	Our firm hires employees who have green awareness/consciousness.	.773
	Our firm develops training programs to increase environmental awareness, skills and expertise of employees.	.742
	Our firm have integrated training to create the emotional involvement of employees in environment management.	.807
	Our firm have green knowledge management (To link environmental education and knowledge to behaviors to develop preventative solutions).	.763
	We use green performance indicators too, in our performance management system and appraisals.	.799
	Our firm sets green targets, goals and responsibilities for managers and employees.	.794
	In our firm, managers are set with objectives on achieving green outcomes included in appraisals.	.795
	In our firm, there are coercive measures in the performance management system for non-compliance or not meeting environment management goals.	.787
	We make green benefits (transport/travel) available rather than giving out pre-paid cards to buy the green products.	.780
	In our firms, there are financial or tax incentives (bicycle loans, use of less polluting cars etc.).	.767
	Our firm has recognition-based rewards in environment management for staff (Eg:public recognition, awards, paid vacations, time off, gift certificates).	.740
	Our company has a clear developmental vision to guide the employees' actions in environment management.	.836
	In our firm, there is a mutual learning climate among employees for green behavior and awareness.	.826
	In our firm, there are a number of formal or informal communication channels to spread green culture in the firm.	.847
	In our firm, employees are involved in quality improvement and problem-solving on green issues (environmental issues).	.812
	We offer practices for employees to participate in environment management, such as newsletters, suggestion schemes, problem-solving groups, low-carbon champions and green action teams.	.832
Sustainable Development	Our firm improved employee or community health and safety.	.735
	Our firm recognized and acted on the need to fund local community initiatives.	.825
	Our firm communicated the firm's environmental impacts and risks to the general public.	.818
	Our firm considered interests of stakeholders in investment decisions by creating a formal dialog.	.677
	Our firm sold waste product for revenue.	.724
	Our firm reduced costs of inputs by the same level of outputs.	.719
	Our firm worked with government officials to protect the company's interest.	.729
	Our firm created spin-off technologies that could be profitably applied to other areas of the firm.	.708
	Our firm reduced energy consumption.	.772
	Our firm reduced impact on animal species and natural habitats.	.760
	Our firm reduced environmental impact by establishing partnerships.	.686
	Our firm reduced purchases of non-renewable materials, chemicals, and components.	.785
	Our firm reduced the use of traditional fuels by substituting some less polluting energy sources.	.731
	Our firm undertook voluntary actions (e.g., actions that are not required by regulations) for environmental restorations.	.758
	Our firm undertook actions for environmental audit, public disclosure, employee training and immunity.	.714
Green Creativity	The members of the green product development project suggest new ways to achieve environmental goals.	.698
	The members of the green product development project propose new green ideas to improve environmental performance.	.709
	The members of the green product development project promote and champion new green ideas to others.	.736
	The members of the green product development project develop adequate plans for the implementation of new green ideas.	.667
	The members of the green product development project would rethink new green ideas.	.722
	The members of the green product development project find creative solutions to environmental problems.	.722

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